

VWCG Unified App Spec v1.1 — Stateless Workspace + High-Value Visual Reporting (Hardened)

Objective

Unify all VWCG tools into **one single application** under one domain while remaining **stateless by design** (no server-side databases, no accounts). Provide a first-class **Save / Load Workspace** system that allows users to fully own, persist, and re-load their strategic work.

Success Criteria (Hard Requirements) - **Workspace round-trip integrity**: save → load → save produces a byte-equivalent file (excluding timestamps). - **Export fidelity**: 100% of required visuals appear in exported PDFs (chart-count verified). - **No-crash guarantee**: corrupted or partial workspace files must fail gracefully with repair options. - **Performance budgets**: - Initial load TTI < 2.5s desktop / < 4s mobile (median device) - Workspace import < 1s for files ≤ 10MB - **Privacy guarantee**: no user-entered content is transmitted or stored server-side.

Primary expansion focus: **visual reporting and exports** as the highest perceived end-user value.

Guiding Principles

- **Stateless by default**: VWCG never stores workspace data server-side.
 - **Client-owned state**: all persistence lives inside a user-controlled workspace file or optional local browser storage.
 - **Deterministic logic**: every score, chart, and recommendation has defined rules and produces repeatable output.
 - **Visuals must explain themselves**: no chart ships without interpretation text.
 - **Failure is expected**: the system must degrade safely under corruption, partial data, or unsupported inputs.
 - **Privacy is structural, not a policy**: architecture prevents data custody by design.
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In Scope (v1.1)

- Single unified SPA (one domain)
- Shared UI shell + tool routing
- Versioned **Workspace State** schema
- Save / Load with validation, repair, and preview
- Standardized visualization system
- Deterministic export pipeline (PDF / JSON / CSV)
- Report Center (per-tool + consolidated reports)

Explicitly Out of Scope

- Server-side databases
 - Authentication / accounts
 - Multi-user collaboration
 - Cloud sync
 - Billing or subscriptions
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Information Architecture

Top-Level Navigation

1. **Dashboard** — workspace overview, tool completion, export status
2. **Tools** — all analysis and planning modules
3. **Report Center** — export generation and history
4. **Workspace** — save, load, validate, repair, clear local data
5. **Help & Trust** — methodology, logic notes, privacy guarantees

Tool Registry (Critical)

Each tool must register itself with: - `toolId` - `schemaNamespace` - `logicVersion` - `requiredVisuals` - `exportCapabilities` - `repairHandler`

No tool may hard-code assumptions about other tools.

Workspace Model

Workspace File

- Extension: `.vwcg`
- Format: JSON (human-readable)
- Max size: **25MB hard cap** (warn at 10MB)
- Integrity: SHA-256 checksum stored in metadata

Workspace Schema (Conceptual)

- `schema_version`
- `workspace_id`
- `created_at`, `updated_at`
- `metadata` (industry, company size, goals)
- `tools` (namespaced by `toolId`)

- `export_log` (stored *inside* workspace, never server-side)
- `integrity` (checksum, validation results)

Non-Negotiable Rules

- Unknown fields are preserved verbatim.
 - Missing tools do not break load.
 - Workspace always loads in **safe mode** first.
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Import / Load Flow (Hardened)

1. User selects `.vwcg` file
2. System performs:
3. JSON validity check
4. checksum verification
5. schema compatibility check
6. User sees **preview panel**:
7. tools present
8. file size
9. last modified
10. warnings/errors
11. User chooses:
12. Load full workspace
13. Load selected tools only
14. Repair & load (non-destructive copy)

Failure Modes (Defined)

- Corrupted JSON → repair attempt or abort
 - Unsupported schema → open in read-only mode
 - Partial tool state → tool-level safe defaults
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Save Flow

- Global Save button always visible
 - Deterministic dirty-state tracking:
 - Tool marks dirty only on data mutation
 - Navigation without mutation does not mark dirty
 - Export generation does **not** mark dirty
 - Optional local autosave (browser only, opt-in)
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Unified UX Standards

Layout Standard

- Left: inputs / steps
- Right: outputs / visuals (sticky on desktop)

Output Contract (All Tools)

Every tool must output: 1. **Scores / status** (where applicable) 2. **Visuals** (defined set) 3. **Interpretation text** (what this means) 4. **Actionable next steps** (prioritized)

Visual Reporting & Export (High-Impact Core)

Export Formats

- **PDF** (primary)
- **JSON** (tool + workspace state)
- **CSV** (tables, timelines, SOPs)

Export Rubric (Mandatory Sections)

1. Executive summary
2. Diagnostic scores
3. Visuals (with captions)
4. Risks & constraints
5. 30 / 60 / 90-day actions
6. Appendix (inputs summary)

Rendering Pipeline (Single Path)

1. Render charts in DOM
2. Rasterize to static images
3. Compose PDF layout
4. Validate chart count
5. Generate file

Export Acceptance Tests

- No blank charts
- Fonts embedded
- Pagination stable
- Works offline
- Dark/light mode safe

Tool Logic Specifications (Expanded)

AI Readiness Assessment

- Scoring model: weighted dimension average (defined weights)
- Normalization: 0–100
- Visuals: radar, priority matrix, maturity ladder

Leadership DNA Radar

- Inputs: 6 leadership dimensions (1–10)
- Gap score = max delta between executives
- Visuals: radar overlays, heatmap

Business Emotional Intelligence

- Dimension scoring with confidence bands
- Supports time-series entries
- Visuals: bar chart + trendline

Vision Canvas

- Pillars: configurable N (default 6)
- Visuals: canvas diagram, KPI dashboard, risk matrix

SWOT Analysis

- Confidence scale: 1–5 per item
- Clustering: keyword similarity + manual merge
- Visuals: quadrant board, distribution chart

SOP Taxonomy Wizard

- Canonical structure: Dept → Category → Process → SOP
- Exports: tree diagram + adjacency CSV

SOP Creation Wizard

- Required blocks: purpose, scope, roles, steps, QC, metrics
- Visuals: flow diagram, checklist

SOP Management Wizard

- Client-side indexing + search

- Review schedule stored locally

90-Day Scaling Roadmap

- Task model: id, phase, dependency, effort, impact, status
- Visuals: progress dashboard, Gantt timeline

Advisor Readiness Assessment (BusinessAdvisors Module)

Purpose

A short, structured diagnostic (target ~7 minutes) to determine whether engaging a business advisor is likely to produce measurable ROI, and to generate an executive-grade report + action roadmap.

Core Outputs

- **Advisor Readiness Score** (0–100) + banding
- **Impact Analysis** (top 5–7 initiatives driving outcomes)
- **Risk Register** (risks + mitigations)
- **ROI Scenarios** (best / likely / worst)
- **6–8 Week Implementation Roadmap** (prioritized)

Visuals (Required)

- Readiness radar (dimension scores)
- Impact bar chart (initiative drivers)
- ROI projection bar chart (baseline, time saved, churn reduced, margin gains, net)

Logic (Deterministic)

- Readiness: weighted dimension score (documented weights)
- Impact: rank initiatives by (impact × confidence) with tie-break by effort
- ROI: scenario table with explicit assumptions stored in workspace (no hidden math)

Privacy & Storage

- Fully compatible with VWCG stateless workspace model.
- No account required in unified app; optional "**Save Workspace**" replaces login.

Exports

- Tool Snapshot PDF (executive-grade)
- JSON tool state
- CSV: risk register + roadmap tasks + ROI assumptions

Cross-Tool Synthesis (User-Controlled)

- Suggestions must record **provenance** (source tool + field)
 - One-click rollback for any auto-generated linkage
 - No silent data propagation Guardrails (Hardened)
 - All synthesis actions recorded as **patch objects** containing:
 - target path
 - source fields
 - transform id + version
 - before value
 - after value
 - base hash (conflict detection)
 - Suggestions enter an explicit review queue
 - No synthesis affects scores or exports until explicitly applied
 - Rollback verifies base hash before apply; conflicts surface to user
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Export DoS & Content Controls

- Max words per chart interpretation: 250
 - Max charts per report: 40
 - Max PDF pages: 40 (overflow moved to appendix or omitted with notice)
 - Chart pixel area capped; SVG node count limited
 - Label text truncated with tooltip where necessary
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Export Pipeline Safeguards

- Single deterministic pipeline: render → rasterize → compose → validate
 - Export job queue:
 - Max 1 concurrent job
 - Cooldown: 10 seconds between jobs
 - Validation checks:
 - no blank charts
 - font embedding verified
 - pagination stable
 - dark/light mode safe
 - Lite export mode auto-triggers when:
 - device memory < threshold
 - rasterization exceeds time budget
 - mobile device detected
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Local Storage & Privacy Controls

- Autosave is opt-in and disabled by default
 - Clear warning: local data is unencrypted
 - One-click **Clear Local Data** control always visible
 - Telemetry (if any) is event-only, payload-free, and opt-in
 - Crash reporting must redact workspace data unless user explicitly consents
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Logic Versioning & Recompute Policy (User-Visible)

- Computed outputs store `computed_under_logic_version`
 - Logic changes never auto-recompute prior results
 - User must explicitly opt in to recompute under new logic
 - Exports must disclose:
 - logic version per tool
 - mixed-version warning if applicable
 - Prior results remain visible for comparison
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Operational Limits (Per Tool)

- Each tool must declare:
 - max entity counts
 - memory budget
 - export cost estimate
 - Export cooldowns prevent repeated stress-triggering actions
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Canonicalization & Round-Trip Integrity

Round-trip integrity is defined as **canonical equivalence**, not raw byte identity.

Canonicalization rules: - Stable JSON key ordering - Stable float rounding rules - Preserved `workspace_id`
- Timestamps may change; all other fields must match canonically - Checksum is recalculated on save

Encryption (Future, Explicitly Non-Default)

- Password-based encryption only
 - Lost password = unrecoverable
 - Clear UX warnings required
 - Not enabled by default
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Appendix A — Workspace Schema v1.0 (Formal)

A1. Design Goals

- Fully reconstruct user work across tools with **no server-side data custody**.
- Support **canonical equivalence** checks for round-trip integrity.
- Preserve forward compatibility safely via `_unknown` quarantine.

A2. Canonical Representation Rules

To enable canonical equivalence and stable exports, the workspace representation must follow: - **Stable key ordering** (lexicographic) when saving. - **Stable numeric rounding** for computed values (e.g., scores rounded to 2 decimals). - **Stable IDs**: `workspace_id` and tool entity IDs never regenerate on load. - **Timestamps may change**: `updated_at` and `last_saved_at` are excluded from equivalence checks. - **Checksum recalculated** on every save.

A3. Top-Level Fields

The workspace MUST contain the following top-level sections.

Identity & Versioning

- `schema_version` (string, required): current schema version (e.g., "1.0.0").
- `workspace_id` (string, required): unique stable identifier.
- `app_build` (string, required): build identifier of the app that produced the file.
- `created_at` (timestamp string, required)
- `updated_at` (timestamp string, required)
- `last_saved_at` (timestamp string, required)

Metadata

- `metadata` (object, required)
- `industry` (string, optional)
- `company_size_band` (string, optional)
- `revenue_band` (string, optional)
- `primary_goals` (array of strings, optional)
- `notes` (string, optional; subject to string length caps)

Tools Container

- `tools` (object, required): map of tool states by `toolId`.
- Each `tools[toolId]` MUST include:
 - `tool_state_version` (string, required)
 - `logic_version` (string, required)
 - `data` (object, required): user inputs + tool-specific structures

- `computed` (object, optional): computed outputs; MUST include `computed_under_logic_version` if present
- `_unknown` (object, optional): quarantined unknown fields preserved byte-for-byte

Export Log

- `export_log` (array, required): stored within the workspace file.
- Each entry includes:
 - `export_id` (string)
 - `timestamp` (string)
 - `type` ("tool_snapshot" | "workspace_executive" | "implementation_pack")
 - `tool_coverage` (array of toolIds)
 - `formats` (array: "pdf" | "json" | "csv")
 - `logic_versions` (map toolId → logicVersion) for auditability

Integrity

- `integrity` (object, required)
- `checksum_sha256` (string, required): corruption detection only
- `size_bytes` (number, required)
- `validation_summary` (object, required)
 - `status` ("ok" | "warn" | "error")
 - `warnings` (array of strings)
 - `errors` (array of strings)

A4. Tool State Minimums (Per Tool)

Every tool's `data` section MUST: - Carry stable entity IDs for list items (SWOT items, SOP nodes, roadmap tasks). - Include `completeness` markers: - `required_fields_present` (boolean) - `defaults_applied` (boolean) - `completeness_score` (0-100)

A5. Limits (Schema-Level)

All fields remain subject to the global hardening limits: - Max depth, max nodes, max string length, max array length - Per-tool size budget - Per-tool entity caps (declared by registry)

Appendix B — Tool Registry Contract (Non-Code, Enforceable)

B1. Registry Purpose

The Tool Registry is the single authority that defines: - What tools exist - What each tool is allowed to store - What each tool must export - What limits each tool must enforce

Registry is **static** (ships with the app build) and validated at build time.

B2. Required Registry Fields (Per Tool)

Each tool must declare:

Identity

- `toolId` (stable unique string)
- `displayName`
- `route` (e.g., `/tools/swot`)

Schema & Logic

- `schemaNamespace` (e.g., `vwcg.tools.swot`)
- `tool_state_version`
- `logic_version`
- `validation_profile` (reference name to the tool's input validation rules)

Limits & Budgets (Hard Requirements)

- `max_entity_counts` (object): caps for each entity type (e.g., SWOT items ≤ 200)
- `max_tool_bytes` (number): must be $\leq$ global per-tool budget
- `memory_budget_hint_mb` (number)
- `export_cost_estimate`:
- `max_charts` (number)
- `max_pages` (number)
- `worst_case_raster_ms` (number)

Visual & Export Requirements

- `required_visuals` (list): exact visual IDs required for compliance
- `export_capabilities` (list): supported exports (PDF/JSON/CSV)
- `report_template_profile` (which PDF layout template to use)

Isolation & Repair

- `namespace_accessor_only` (boolean, must be true): tool must only receive scoped access to its namespace
- `repair_handler_profile` (string): identifies the repair behaviors allowed
- `repair_mutation_scope` (must be `self_only`)

B3. Enforcement Rules

- Tools cannot access peer namespaces.
- Registry caps override any UI or tool code.
- If a tool exceeds limits, the tool must:

- stop accepting new entities
 - show “limit reached” messaging
 - maintain export compliance
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Appendix C — Export Pipeline Failure Injection Test Plan

C1. Test Philosophy

Exports fail in real life due to: content extremes, browser constraints, fonts, SVG complexity, and race conditions. The system must handle these predictably.

C2. Core Test Suites

Suite 1 — Content Extremes

- **Max label length:** chart labels at max allowed length; verify truncation + tooltip.
- **Max interpretation text:** hit per-chart and total report limits; verify overflow behavior.
- **High entity counts:** near-cap SWOT/SOP/Roadmap; verify performance + export correctness.

Suite 2 — SVG / Raster Stress

- Dense SVG (many nodes) within allowed limits; verify raster time budget triggers lite mode.
- Oversized chart surfaces attempt; verify pixel-area cap enforcement.

Suite 3 — Pagination & Layout Stability

- Long tables (CSV-backed) rendered into PDF; verify stable pagination.
- Mixed portrait/landscape elements; verify no clipping.
- Dark mode enabled; verify exports remain readable.

Suite 4 — Offline & Asset Availability

- Export immediately after initial load with airplane mode; verify:
- PDF library available offline
- required fonts available offline
- images embedded correctly

Suite 5 — Failure & Recovery

- Force export interruption mid-stream (simulate exception):
- job reports failure
- export log records failure state
- user can retry without corruption

- Storage quota exceeded during export log update: app warns and continues export without crashing.

Suite 6 — Concurrency / Race

- Attempt multiple exports rapidly:
- verify queue behavior
- verify cooldown enforcement
- Trigger Clear Local Data during export:
- verify export completes or cleanly aborts
- verify no partial writes

C3. Export Acceptance Checks (Automated Where Possible)

- Chart-count validation equals registry required visuals
 - No blank chart placeholders
 - Font embedding verified
 - File size within expected bounds
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Appendix D — Safe Mode UX Copy (Explicit)

D1. Safe Mode Banner (Top of Screen)

Safe Mode: Preview Only This workspace file is being previewed safely. No tools are running, and no charts are rendered yet.

What you can do now: - Review what's inside the workspace - Choose which tools to load - Run a strict load or a repair load

D2. Warnings Panel Copy

Integrity Check

Integrity check passed (corruption detection). This does not guarantee the file is safe or trustworthy. VWCG will treat all content as untrusted until you load it.

Defaults Applied

Some tool data is incomplete. If you load this workspace, certain tools may use safe defaults. Any scores based on defaults will be clearly marked.

Unsupported / Partial Sections

Some sections are **partial or unsupported**. You can still load supported tools. Unsupported data will be preserved under a protected `_unknown` section.

D3. Load Options Copy

Strict Load (Recommended)

Loads the workspace without modifying any data. If something is invalid, you'll see a clear error report.

Repair Load (Creates a Copy)

Attempts a safe repair and generates a new workspace file with a repair report. Your original file is never changed.

D4. Safe Mode Exit Copy

When you click **Load Selected Tools**, VWCG will activate only those tools. You can return to Safe Mode at any time.

Appendix E — Validation Profiles (Per Tool, Deterministic)

E0. Validation Philosophy

Validation must be deterministic, fast, and failure-aware.

Validation levels: - **L0: Structural** (handled by global workspace constraints) - **L1: Syntactic** (required fields, types, bounds) - **L2: Semantic** (cross-field consistency) - **L3: Diagnostic Integrity** (completeness + defaults applied flags)

Rules - Validation must never execute untrusted HTML or SVG. - Validation must fail fast and return actionable error codes. - If a tool is incomplete, it must not silently compute "green" results.

Each profile below defines: - Required fields - Allowed ranges - Semantic invariants - Completeness criteria - Default behavior - Error/warning codes

E1. AI Readiness Assessment — validation_profile: airreadiness_v1

Required Fields (L1)

- `data.dimensions` (array, 1-12 items)
- each dimension:
 - `id` (string)
 - `name` (string, ≤ 120)
 - `score` (number, 0-100)
 - `weight` (number, 0-1)
- `data.context.industry` (optional)

Range Rules (L1)

- Sum of weights must be within **0.99-1.01**

Semantic Invariants (L2)

- Dimension IDs unique
- If `score` missing for any dimension → tool incomplete

Diagnostic Integrity (L3)

- `defaults_applied = true` if any missing score filled
- `completeness_score = 100` only if all scores provided

Defaults

- Missing score defaults to **50** but forces:
- Data Incomplete banner
- Confidence downgrade

Codes

- `AIR-REQ-001` missing dimensions
 - `AIR-WGT-001` weights sum invalid
 - `AIR-DUP-001` duplicate dimension id
 - `AIR-INC-001` incomplete scores (warn)
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E2. Leadership DNA Radar — validation_profile: leadership_radar_v1

Required Fields (L1)

- `data.executives` (array, 1-10)
- each exec:
 - `id` (string)
 - `label` (string, ≤ 80)
 - `dimensions` (array, exactly 6)
 - each:
 - `id` (string)
 - `score` (number, 1-10)

Semantic Invariants (L2)

- All execs share the same 6 dimension IDs

Diagnostic Integrity (L3)

- If any exec missing a dimension score → incomplete

Defaults

- No silent defaults; missing scores block computation.

Codes

- `LDR-REQ-001` no executives
- `LDR-DIM-001` wrong dimension count
- `LDR-MAP-001` dimension id mismatch across execs
- `LDR-MIS-001` missing score (error)

E3. Business Emotional Intelligence — validation_profile: bei_v1

Required Fields (L1)

- `data.entries` (array, 1-365)
- each entry:
 - `id` (string)
 - `date` (YYYY-MM-DD)
 - `dimensions` (array, exactly 6)
 - each:
 - `id` (string)

- `score` (number, 1–10)
- `confidence` (number, 1–5)

Semantic Invariants (L2)

- Dates unique per day
- Trend visuals computed only if ≥ 2 entries

Defaults

- No defaults for missing scores; block.

Codes

- `BEI-REQ-001` no entries
- `BEI-DATE-001` invalid date
- `BEI-DUP-001` duplicate date
- `BEI-MIS-001` missing score

E4. Vision Canvas — `validation_profile: vision_canvas_v1`

Required Fields (L1)

- `data.pillars` (array, 3–12)
- each pillar:
 - `id` (string)
 - `title` (string, ≤ 120)
 - `statement` (string, ≤ 2000)
- `data.kpis` (array, 0–50)
- each kpi:
 - `id` (string)
 - `name` (string, ≤ 120)
 - `definition` (string, ≤ 500)
 - `target` (string, ≤ 120)

Semantic Invariants (L2)

- Pillar IDs unique
- KPI IDs unique

Diagnostic Integrity (L3)

- Incomplete if any pillar missing title or statement

Defaults

- No defaults; incomplete blocks canvas export until fixed or user accepts incomplete export.

Codes

- VIS-REQ-001 missing pillars
 - VIS-LIM-001 pillar count out of bounds
 - VIS-MIS-001 missing pillar fields
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E5. SWOT Analysis — validation_profile: swot_v1

Required Fields (L1)

- data.items (array, 1-800)
- each item:
 - id (string)
 - quadrant (enum: S|W|O|T)
 - text (string, ≤ 500)
 - confidence (number, 1-5)

Semantic Invariants (L2)

- Each quadrant must have ≥ 1 item for complete status

Diagnostic Integrity (L3)

- If any quadrant empty → incomplete banner + export includes disclaimer

Defaults

- Confidence defaults to 3 only if user explicitly chooses “auto-fill confidence”

Codes

- SWT-REQ-001 no items
 - SWT-QDR-001 invalid quadrant
 - SWT-INC-001 missing quadrant items (warn)
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E6. SOP Taxonomy Wizard — validation_profile: sop_taxonomy_v1

Required Fields (L1)

- data.nodes (array, 1-10,000)
- each node:
 - id (string)
 - type (enum: dept|category|process|sop)

- `name` (string, ≤ 120)
- `parent_id` (string | null)

Semantic Invariants (L2)

- Exactly one root (`parent_id = null`)
- No cycles (DAG/tree enforcement)
- Depth ≤ 6

Diagnostic Integrity (L3)

- If cycles detected $\rightarrow$ error, block export

Codes

- `TAX-REQ-001` no nodes
- `TAX-ROOT-001` invalid root count
- `TAX-CYC-001` cycle detected
- `TAX-DPT-001` depth exceeded

E7. SOP Creation Wizard — `validation_profile: sop_create_v1`

Required Fields (L1)

- `data.sop` object:
 - `id` (string)
 - `title` (string, ≤ 120)
 - `purpose` (string, ≤ 800)
 - `scope` (string, ≤ 800)
 - `roles` (array, 1-20)
 - `steps` (array, 1-300)
 - each step:
 - `id` (string)
 - `text` (string, ≤ 800)
 - `qc_check` (optional, ≤ 300)

Semantic Invariants (L2)

- Step IDs unique
- Steps in strict order (no missing indices)

Diagnostic Integrity (L3)

- If missing purpose/scope/steps $\rightarrow$ incomplete

Defaults

- No silent defaults; tool prompts user to fill required blocks.

Codes

- `SOP-REQ-001` missing required blocks
- `SOP-STP-001` no steps

E8. SOP Management Wizard — `validation_profile:` `sop_manage_v1`

Required Fields (L1)

- `data.library` (array, 0-5000)
- each sop summary:
 - `id` (string)
 - `title` (string)
 - `next_review_date` (YYYY-MM-DD)
 - `status` (enum: draft|active|deprecated)

Semantic Invariants (L2)

- Dates validated in user timezone
- Review dates must be $\geq$ created date

Failure Modes

- Browser quota exceeded:
 - warn user
 - disable autosave
 - allow export-only mode

Codes

- `LIB-DATE-001` invalid review date
- `LIB-QUO-001` storage quota exceeded (warn)

E9. 90-Day Scaling Roadmap — `validation_profile:` `roadmap_90_v1`

Required Fields (L1)

- `data.tasks` (array, 1-5000)

- each task:
 - `id` (string)
 - `phase` (enum: 1|2|3)
 - `title` (string, ≤ 160)
 - `status` (enum: not_started|in_progress|blocked|done)
 - `effort` (number, 1-5)
 - `impact` (number, 1-5)
 - `dependencies` (array of task ids, ≤ 50)

Semantic Invariants (L2)

- No dependency cycles
- Dependencies must exist

Diagnostic Integrity (L3)

- If cycles detected → block timeline export

Codes

- `RDM-REQ-001` no tasks
- `RDM-DEP-001` missing dependency
- `RDM-CYC-001` dependency cycle

Appendix F — Required Sanitization & Rendering Rules

F1. Global Rule

All user-provided and workspace-provided text MUST be treated as **untrusted**.

F2. Text Rendering

- Render text as **plain text only**
- Never render user content via HTML injection APIs
- Escape all special characters when inserted into DOM

F3. SVG / Chart Safety

- Chart labels/tooltips must be sanitized and length-limited
- Cap SVG node count and chart pixel area (per export safeguards)
- Disallow external references in SVG (no remote URLs)

F4. PDF Composition Safety

- Truncate or wrap long strings
 - Enforce page limits and appendix overflow rules
 - If constraints are exceeded, export must succeed in lite mode with explicit disclaimer
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Appendix G — Multi-Tab & Local Storage Conflict Handling

G1. Multi-Tab Detection

- Store an active tab session id in local storage
- If another tab opens the same workspace:
 - show conflict banner
 - allow read-only mode in secondary tabs

G2. Autosave Mutex

- Autosave operations must acquire a mutex
 - Clear Local Data must acquire the same mutex
 - If Clear Local Data is triggered:
 - autosave pauses
 - user receives confirmation of completion
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Appendix H — Offline Asset Requirements

To satisfy offline export requirements: - PDF generation libraries must be bundled with the app (no CDN dependency) - Required fonts must be bundled and cached at initial load - Export pipeline must not fetch remote assets during generation

Appendix I — Export Version Disclosure Requirements

All exports must include: - Tool logic version per tool - “Mixed logic versions” warning banner if applicable - Statement: “Scores are computed under the listed logic versions; recompute requires explicit user action.”

Appendix J — Advisor Readiness Assessment (BusinessAdvisors) Registry + Validation Addendum

J1. Tool Registry Entry (Declarative)

- `toolId`: `advisor_readiness`
- `displayName`: `Advisor Readiness Assessment`
- `route`: `/tools/advisor-readiness`
- `schemaNamespace`: `vwcg.tools.advisor_readiness`
- `tool_state_version`: `1.0.0`
- `logic_version`: `1.0.0`
- `validation_profile`: `advisor_readiness_v1`
- `max_entity_counts`:
- `questions`: `120`
- `initiatives`: `50`
- `risk_items`: `200`
- `roi_assumptions`: `80`
- `roadmap_tasks`: `200`
- `max_tool_bytes`: `5,000,000`
- `memory_budget_hint_mb`: `64`
- `export_cost_estimate`:
- `max_charts`: `6`
- `max_pages`: `18`
- `worst_case_raster_ms`: `2500`
- `required_visuals`:
- `readiness_radar`
- `impact_bar`
- `roi_projection_bar`
- `export_capabilities`: `[pdf, json, csv]`
- `report_template_profile`: `executive_diagnostic`
- `namespace_accessor_only`: `true`
- `repair_handler_profile`: `self_only_minimal`
- `repair_mutation_scope`: `self_only`

J2. Validation Profile — `advisor_readiness_v1`

Required Fields (L1)

- `data.responses` (array, 10-120)
- each response:
 - `id` (string)
 - `dimension` (string)

- `question` (string, ≤ 300)
- `answer` (enum: 1-5 OR boolean OR short string ≤ 300 , per question type)
- `confidence` (number, 1-5)
- `data.dimensions` (array, 3-12)
- each:
 - `id` (string)
 - `name` (string, ≤ 120)
 - `weight` (number, 0-1)

ROI Assumptions (L1)

- `data.roi` object:
- `baseline` (number, ≥ 0)
- `drivers` (array, 1-10)
 - each driver:
 - `id` (string)
 - `name` (enum: baseline | time_saved | churn_reduced | margin_gains | net | other)
 - `best` (number)
 - `likely` (number)
 - `worst` (number)
 - `assumption_note` (string, ≤ 500)

Risk Register (L1)

- `data.risks` (array, 0-200)
- each risk:
 - `id` (string)
 - `risk` (string, ≤ 240)
 - `severity` (number, 1-5)
 - `likelihood` (number, 1-5)
 - `mitigation` (string, ≤ 500)

Roadmap (L1)

- `data.roadmap_tasks` (array, 0-200)
- each task:
 - `id` (string)
 - `week` (number, 1-8)
 - `title` (string, ≤ 160)
 - `owner` (string, ≤ 80)
 - `status` (enum: not_started | in_progress | blocked | done)

Semantic Invariants (L2)

- Sum of dimension weights within **0.99-1.01**
- Every response maps to a known dimension

- ROI scenario values must satisfy: `best >= likely >= worst` OR explicitly flagged with `non_monotonic_ok=true`

Diagnostic Integrity (L3)

- If `responses` missing for any required dimension → incomplete banner
- If ROI assumptions missing → exports allowed but must label ROI section “Not Provided”
- Defaults are permitted only when user explicitly selects “Generate baseline assumptions”

Codes

- `ADR-REQ-001` missing responses
- `ADR-DIM-001` dimension weight invalid
- `ADR-MAP-001` response dimension unknown
- `ADR-ROI-001` ROI drivers missing
- `ADR-ROI-002` ROI ordering invalid (warn or error depending on flag)

J3. Export Requirements (Advisor Readiness)

- PDF must include:
 - Advisor Readiness Score + band
 - Readiness radar with interpretation
 - Impact bar chart with top 5–7 initiatives
 - Risk register summary (top risks) + full appendix table if long
 - ROI scenarios (best/likely/worst) + assumptions table
 - 6–8 week roadmap (week-by-week)
 - Logic version disclosure block (Appendix I)

J4. Workspace Integration (No Accounts)

- Unified VWCG app must not require login.
- “Save Workspace” replaces account persistence:
- Users can download `.vwcg`
- Users can optionally enable local autosave (opt-in)
- If a separate product later adds accounts/cloud sync, it must remain optional and must not change local-first defaults.